

Chapter 4 — Forest Society and Colonialism

The one-line point: Colonial rule did not merely **take** the forest — it **redefined** it. A forest that had been a place people lived *inside*, drawing food, medicine, fodder, rope and roofing from hundreds of species, was reclassified as a **timber estate** growing one species in straight rows. Everything villagers had always done in it — cutting wood, grazing cattle, gathering fruit, hunting, farming — **became a crime on the same day**.

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Syllabus Classification

Field	Value
Subject	History → Modern India + Environmental history
Sub-discipline	Ecological history · colonial land policy · tribal history
Period	c. 1700 – 1940s (core: 1850–1930)
Prelims (GS Paper I & III)	★ ★ ★ The highest-yield chapter in this book. Dietrich Brandis, the Indian Forest Service 1864, the Forest Acts of 1865/1878/1927, reserved–protected–village forests, FRI Dehradun 1906, shifting-cultivation terms, the Bastar rebellion — all directly askable
Mains (GS Paper I)	★ ★ ★ <i>"colonization... their forms and effect on the society"; tribal movements; the ecological cost of colonial rule</i>
Also feeds	★ ★ ★ ★ GS3 Environment (Forest Rights Act 2006, Joint Forest Management, community conservation, sacred groves, monoculture vs biodiversity), GS3 Agriculture (shifting cultivation), GS2 (rights of forest dwellers, PESA), Indian Society ★ ★ ★ (tribal displacement), Essay ★ ★ ★ (development and dispossession; whose knowledge counts)
Weightage	★ ★ ★ ★ — this chapter pays back in three different papers
Overlaps with	★ ★ ★ Class 8 Ch.4 (<i>Tribals, Dikus and the Vision of a Golden Age</i> — the same policy seen from the tribal side; Birsa Munda appears in both), Ch.5 of this book (pastoralists), Class 12 Themes III

Learning Outcomes

By the end of this chapter you should be able to:

- ☐ **List the everyday things that come from forests** — NCERT's own opening exercise. ★
- ☐ **Give the global deforestation figure** for 1700–1995. ★ ★
- ☐ **Explain the FOUR causes of colonial deforestation** with figures for each. ★ ★ ★ ★
- ☐ **Explain *terra nullius*** and why "uncultivated" did not mean "uninhabited". ★ ★ ★
- ☐ **Name Dietrich Brandis** and everything he set up, with dates. ★ ★ ★ ★
- ☐ **Define scientific forestry** and say why ecologists dispute the name. ★ ★ ★ ★
- ☐ **Distinguish reserved, protected and village forests.** ★ ★ ★ ★
- ☐ **Describe shifting cultivation** and list its names across the world. ★ ★ ★
- ☐ **State the THREE reasons the British banned it.** ★ ★ ★ ★
- ☐ **Explain the contradiction in colonial hunting policy**, with the numbers. ★ ★ ★
- ☐ **Explain who gained and who lost** from the new forest trade. ★ ★ ★

- ☐ **Narrate the Bastar rebellion of 1910** — causes, method, leaders, outcome. ★★★★★
- ☐ **Compare Bastar with Java** — this is asked directly. ★★★★★
- ☐ **Explain the *blandongdiensten* system and Samin's challenge.** ★★★
- ☐ **Explain why wars destroy forests.** ★★★
- ☐ **Describe the post-1980s shift** toward community conservation. ★★★★★

Start Here: Look Around the Room

NCERT opens by asking you to identify everything around you that comes from a forest:

the PAPER in this book · DESKS and TABLES · DOORS and WINDOWS
 the DYES that colour your clothes · SPICES in your food
 the CELLOPHANE wrapper of a toffee · TENDU LEAF in bidis
 GUM · HONEY · COFFEE · TEA · RUBBER
 the OIL IN CHOCOLATES (from SAL SEEDS)
 TANNIN (to convert skins and hides into leather)
 HERBS and ROOTS for medicine
 BAMBOO · FUELWOOD · GRASS · CHARCOAL · PACKAGING
 FRUITS · FLOWERS · ANIMALS · BIRDS

★ **The diversity figure:** *"In the **Amazon** forests or in the **Western Ghats**, it is possible to find as many as **500 different plant species in ONE forest patch.**"*

★★★ **The loss figure — memorise it:** *"Between **1700 and 1995**, the period of **industrialisation**, **13.9 million sq km of forest** — **9.3 per cent of the world's total area** — was cleared for **industrial uses, cultivation, pastures and fuelwood.**"*

Deforestation = the disappearance of forests. NCERT's framing sentence: *"Deforestation is **not a recent problem**. The process began many centuries ago; **but under colonial rule it became more SYSTEMATIC and EXTENSIVE.**"*

 **Note the two adjectives — *systematic* and *extensive*.** They are the whole argument of the chapter, and they are the right words to use in an answer. **Colonialism did not invent clearing; it industrialised it and gave it a bureaucracy.**

Why Deforestation? — The Four Causes

1.1 "Land to be improved" — agricultural expansion

Fact	Figure
★★ India's landmass under cultivation, 1600	approximately one-sixth
★★ Now	about half
★★★ Cultivated area rose, 1880–1920	by 6.7 million hectares

The two colonial-period drivers:

FIRST -- COMMERCIAL CROPS

"The British DIRECTLY ENCOURAGED the production of commercial crops like JUTE, SUGAR, WHEAT and COTTON."

WHY? Demand rose in 19th-century Europe, where

- FOODGRAINS were needed to feed the GROWING URBAN POPULATION
- RAW MATERIALS were required for INDUSTRIAL PRODUCTION

SECOND -- THE OFFICIAL VIEW OF FORESTS

"In the early nineteenth century, the colonial state thought that forests were UNPRODUCTIVE. They were considered to be WILDERNESS that had to be BROUGHT UNDER CULTIVATION so that the land could yield AGRICULTURAL PRODUCTS AND REVENUE, and ENHANCE THE INCOME OF THE STATE."

★★★★ NCERT's warning sentence — quote it: **"We always see the expansion of cultivation as a sign of progress. But we should not forget that for land to be brought under the plough, forests have to be cleared."**

🌐 Box 1 + Source A — *terra nullius*, and the argument that justified conquest

Box 1: **"The absence of cultivation in a place does not mean the land was uninhabited.** In Australia, when the white settlers landed, they claimed that the continent was **empty — *terra nullius***. In fact, they were guided through the landscape by aboriginal tracks, and led by aboriginal guides. The different aboriginal communities in Australia had **clearly demarcated territories**. The ★ Ngarrindjeri people plotted their land along the **symbolic body of the first ancestor, Ngurunderi**. This land included **five different environments — salt water, riverine tracts, lakes, bush and desert plains — which satisfied different socio-economic needs.**"

Source A: **"The idea that uncultivated land had to be taken over and improved was popular with colonisers everywhere in the world. It was an argument that justified conquest."** The American writer **Richard Harding on Honduras, 1896:** *"There is no more interesting question... than that of what is to be done with the world's land which is lying unimproved; whether it shall go to the great power that is willing to turn it to account, or remain with its original owner, who fails to understand its value. The Central Americans are like a gang of semi-barbarians in a beautifully furnished house, of which they can understand neither its possibilities of comfort nor its use."* ★ **Three years later the American-owned UNITED FRUIT COMPANY was founded**, growing bananas on an industrial scale in Central America, and acquiring such power over those governments that they became known as ★★ **"BANANA REPUBLICS"**.

★★★★ **This is one of the most transferable ideas in the whole NCERT series.** The colonial move is always the same: **declare the land empty or wasted → declare the inhabitants incapable of valuing it → conclude that taking it is improvement rather than theft.** You will meet exactly this logic again in **Chapter 5** (the *Waste Land Rules*), in **Class 8 Ch.3** (land the British called "waste"), and in every modern argument about "unproductive" land. **Note also the sequence in Honduras: the rhetoric came first, the company three years later.**

🚂 1.2 "Sleepers on the tracks" — ships and railways

First, ships:

"By the early nineteenth century, OAK FORESTS IN ENGLAND WERE DISAPPEARING." -> a problem of TIMBER SUPPLY FOR THE ROYAL NAVY.

"How could English ships be built without a regular supply of strong and durable timber? HOW COULD IMPERIAL POWER BE PROTECTED AND MAINTAINED WITHOUT SHIPS?"

BY THE 1820s -- SEARCH PARTIES were sent to explore the FOREST RESOURCES OF INDIA.

"Within a DECADE, trees were being felled ON A MASSIVE SCALE and vast quantities of timber were being EXPORTED FROM INDIA."

Then, railways — from the 1850s:

Fact	Figure
Why railways mattered	"essential for colonial trade and for the movement of imperial troops "
Two demands	wood as fuel to run locomotives, and ★ sleepers to hold the tracks together
★★★ Sleepers per mile of track	between 1,760 and 2,000
★★ Track laid by 1890	about 25,500 km
★★ Track by 1946	over 765,000 km
★★ Madras Presidency alone, as early as the 1850s	35,000 trees cut annually for sleepers
★★★ The mechanism of destruction	*"The government gave out contracts to individuals to supply the required quantities. These contractors began cutting trees indiscriminately. Forests around the railway tracks fast started disappearing. "*

Word: ★ Sleepers — wooden planks laid across railway tracks; they hold the tracks in position.

Source B (E.P. Stebbing, *The Forests of India*, 1923) puts one line's demand on the record: the **Indus Valley Railway from Multan to Sukkur**, nearly **300 miles**, at 2,000 sleepers a mile = **600,000 sleepers**, upwards of **2,000,000 cubic feet**; plus wood fuel for the locomotives at **219,000 maunds a year**; plus fuel for brick-burning. And the **Northern State Railway from Lahore to Multan** was estimated to need **2,200,000 sleepers**.

🎯 **Do NCERT's arithmetic once and you will never forget it.** One average tree yields **3 to 5 sleepers**. At **1,760–2,000 sleepers per mile**, one mile of broad-gauge track consumed roughly **350 to 650 trees** — before counting the fuel the trains then burned. Multiply by **765,000 km**. **The railway map of India is also a deforestation map.**

★★★ **And note the cruelty recorded in the caption to Fig. 3 (Singhbhum, Chhotanagpur, 1897):**
*"Adivasis were hired by the forest department to cut trees, and make smooth planks which would serve as sleepers for the railways. At the same time, **THEY WERE NOT ALLOWED TO CUT THESE TREES TO BUILD THEIR OWN HOUSES.**"* That single sentence contains the entire chapter.

1.3 Plantations

"Large areas of natural forests were also cleared to make way for ★ **TEA, COFFEE and RUBBER plantations** to meet **Europe's growing need for these commodities**. The colonial government **took over the forests, and gave vast areas to EUROPEAN PLANTERS AT CHEAP RATES**. These areas were **enclosed and cleared of forests**, and planted with tea or coffee."

🎯 Four causes, then, in one line each — this is the shape of the answer to NCERT's Question 3: (i) **agricultural expansion** for commercial crops and revenue; (ii) **shipbuilding** for the Royal Navy after England's oaks ran out; (iii) **railways** — sleepers and fuel, with indiscriminate contractors; (iv) **plantations** — tea, coffee, rubber on cheap land given to European planters.

2 The Rise of Commercial Forestry

👤 Dietrich Brandis and the machinery he built — ★★★★★ pure Prelims gold

"The British were worried that **the use of forests by local people and the reckless felling of trees by traders** would destroy forests. So they decided to invite a **German expert, DIETRICH BRANDIS**, for advice, and made him the ★ **FIRST INSPECTOR GENERAL OF FORESTS IN INDIA**."

What Brandis concluded was needed:

1. "A PROPER SYSTEM had to be introduced to MANAGE the forests"
2. "PEOPLE HAD TO BE TRAINED in the SCIENCE OF CONSERVATION"
3. "This system would need LEGAL SANCTION"
4. "RULES about the use of forest resources had to be framed"
5. "FELLING OF TREES AND GRAZING had to be RESTRICTED so that forests could be PRESERVED FOR TIMBER PRODUCTION"
6. "Anybody who cut trees without following the system had to be PUNISHED"

Year	Institution / Act
★★★★ 1864	Indian Forest Service set up by Brandis
★★★★ 1865	Indian Forest Act — Brandis helped formulate it
★★★★ 1906	Imperial Forest Research Institute set up at DEHRADUN (the Imperial Forest School there was the first forestry school inaugurated in the British Empire)

★★★★ Read point 5 again. Grazing and felling were restricted "so that forests could be **preserved for TIMBER PRODUCTION**." Conservation here means **conserving the supply of timber for the state — not conserving the forest as an ecosystem**. Getting this distinction right is what separates a good answer from a vague one.

🌲 "Scientific forestry" — and why the name is disputed

Word: ★★ Scientific forestry — "a system of cutting trees controlled by the forest department, in which **OLD TREES ARE CUT AND NEW ONES PLANTED**."

★★★★ NCERT's own verdict, in the text: *"The system they taught here was called 'scientific forestry'. Many people now, including ecologists, feel that this system is NOT SCIENTIFIC AT ALL."*

THE NATURAL FOREST

"lots of DIFFERENT TYPES of trees"
up to 500 plant species in one patch (Amazon, Western Ghats)
different heights, dense canopy

yields food, fodder, medicine, rope, fuel, oil, dye, fibre

THE PLANTATION

"ONE TYPE of tree planted IN STRAIGHT ROWS"
uniform height, uniform age
(NCERT's picture: a managed POPLAR forest in Tuscany -- "good mainly for timber. They are NOT used for leaves, fruit or other products.")
yields TIMBER

THE PROCEDURE

"Forest officials SURVEYED the forests, ESTIMATED the area under different types of trees, and made WORKING PLANS for forest management. They planned HOW MUCH of the plantation area to CUT EVERY YEAR. The area cut was then to be REPLANTED so that it was ready to be CUT AGAIN in some years."

★★★★ Why ecologists object — state this in your own answer: a monoculture of uniform age is not a forest. It has **no understorey, little biodiversity, poor soil regeneration and high vulnerability to pest and fire**, and it **produces nothing the surrounding community can eat, feed to cattle, or make into rope**. The system is *systematic*, not *scientific*: it optimises **one output** (timber volume) and calls the other outputs waste.

⚖️ The Forest Act and its three categories — ★★★★★ certain Prelims material

THE INDIAN FOREST ACT, 1865
amended TWICE: 1878 and 1927

THE 1878 ACT DIVIDED FORESTS INTO THREE CATEGORIES:

- | |
|---|
| 1. RESERVED FORESTS -- "THE BEST FORESTS"
"Villagers could take NOTHING from these forests, EVEN FOR THEIR OWN USE." |
| 2. PROTECTED FORESTS |
| 3. VILLAGE FORESTS

"For HOUSE BUILDING or FUEL, they could take wood from PROTECTED or VILLAGE forests." |

🔗 Carry this straight into GS3. The reserved/protected distinction still exists in Indian forest law, and the whole architecture of the **Forest Rights Act, 2006** exists to undo what the 1878 Act did — it restores to forest dwellers the very rights (residence, minor forest produce, grazing, community management) that this Act extinguished. **Say that in an answer and you have connected 1878 to 2006 in one sentence.**

🌿 2.1 How the lives of people were affected

The core conflict, stated as NCERT states it:

★★★★★ *"Foresters and villagers had very different ideas of what a good forest should look like. Villagers wanted forests with a **MIXTURE OF SPECIES** to satisfy different needs — fuel, fodder, leaves. The forest department on the other hand wanted **trees which were suitable for building ships or railways**. They needed trees that could provide **hard wood**, and were tall and straight. So particular species like **TEAK** and **SAL** were promoted and others were cut."

What a forest actually supplies a household — learn a few of these, they make an answer concrete:

Product	Use
Fruits and tubers	"nutritious to eat, <i>especially during the monsoons before the harvest has come in</i> "
Herbs	medicine
Wood	agricultural implements like yokes and ploughs
Bamboo	" <i>excellent fences</i> ", baskets and umbrellas
A dried scooped-out gourd	a portable water bottle
Leaves	" <i>stitched together to make disposable plates and cups</i> "
★ Siadi (<i>Bauhinia vahlii</i>) creeper	ropes
★ Thorny bark of the semur (silk-cotton) tree	to grate vegetables
★ Mahua (<i>Madhuca indica</i>)	fruit pressed for oil for cooking and to light lamps ; flowers can be eaten or made into alcohol; seeds make oil . " <i>Villagers wake up before dawn and go to the forest to collect the mahua flowers which have fallen on the forest floor. Mahua trees are precious</i> "
★ Tendu leaves	" <i>a major source of income for many people living in forests. Each bundle contains approximately 50 leaves, and if a person works very hard they can perhaps collect as many as 100 bundles in a day. Women, children and old men are the main collectors</i> "

And then the Act:

★★★★★ *"The Forest Act meant **severe hardship for villagers across the country**. After the Act, **ALL their everyday practices — cutting wood for their houses, grazing their cattle, collecting fruits and roots, hunting and fishing — BECAME ILLEGAL**. People were now forced to **STEAL WOOD** from the forests, and if they were caught, they were at the mercy of the forest guards who would take **BRIBES** from them. Women who collected fuelwood were especially worried. It was also common for **police constables and forest guards to harass people by demanding FREE FOOD from them.**"

★★★★★ The sentence to build an answer around: "**all their everyday practices... became illegal.**" Nothing about the villagers' behaviour changed. **The law changed, and overnight an entire population became criminals in the place they lived.** Note also the second-order effect NCERT flags: **an unenforceable law does not stop the activity — it converts it into a bribe.** The forest guard's power was the real change.

🔥 2.2 How forest rules affected cultivation — shifting cultivation

The names — NCERT lists them, and a Prelims question can be built on any of them:

SHIFTING CULTIVATION / SWIDDEN AGRICULTURE

"a TRADITIONAL agricultural practice in many parts of ASIA, AFRICA and SOUTH AMERICA"

ABROAD: LADING -- Southeast Asia
MILPA -- Central America
CHITEMENE
or TAVY -- Africa
CHENA -- Sri Lanka

IN INDIA: DHYA · PENDA · BEWAR · NEVAD ·
JHUM · PODU · KHANDAD · KUMRI

How it works:

1. Parts of the forest are CUT AND BURNT IN ROTATION
2. SEEDS ARE SOWN IN THE ASHES after the FIRST MONSOON RAINS
3. The crop is HARVESTED BY OCTOBER–NOVEMBER
4. Such plots are cultivated for A COUPLE OF YEARS ...
5. ... then LEFT FALLOW for 12 TO 18 YEARS
"for the FOREST TO GROW BACK"
6. A MIXTURE OF CROPS is grown:
central India and Africa -> MILLETS
Brazil -> MANIOC
other parts of Latin America -> MAIZE and BEANS

★★★★ Why the British banned it — THREE reasons, and you must give all three:

#	Reason (NCERT's wording)
1	"European foresters regarded this practice as harmful for the forests . They felt that land which was used for cultivation every few years could not grow trees for railway timber "
2	"When a forest was burnt, there was the added danger of the flames spreading and burning valuable timber "
★ 3	*"Shifting cultivation also made it harder for the government to CALCULATE TAXES ."*

The consequences: "Therefore, the government decided to **ban shifting cultivation**. As a result, **many communities were FORCIBLY DISPLACED from their homes in the forests**. Some had to change occupations, while **some resisted through large and small rebellions**."

★★★★ Notice that all three reasons are about the STATE'S needs, not the forest's health — timber for railways, protection of that timber, and legibility for taxation. The **12–18 year fallow** was in fact a **regeneration cycle**; the practice was sustainable at the population densities involved. **The objection was never that it destroyed forests; it was that it did not produce what the state wanted, in a form the state could count.**

🔗 **Class 8 Ch.4 link, worth writing in the margin:** there NCERT told you that after widespread protest "the British ultimately allowed jhum cultivators in the north-east the right to continue shifting cultivation in some parts of the forest" — because the attempt to settle them **failed on its own terms**: plough cultivation gave **poor yields where water was scarce and soil dry**. The policy was reversed not out of justice but because it did not work.

🦁 2.3 Who could hunt? — the contradiction

BEFORE THE FOREST LAWS

"Many people who lived in or near forests had survived by HUNTING DEER, PARTRIDGES and a variety of SMALL ANIMALS."

AFTER

"This CUSTOMARY PRACTICE WAS PROHIBITED by the forest laws. Those who were caught hunting were now punished for POACHING."

MEANWHILE

"While the forest laws DEPRIVED PEOPLE of their customary rights to hunt, HUNTING OF BIG GAME BECAME A SPORT."

The pre-colonial background: *"In India, hunting of tigers and other animals had been **part of the culture of the court and nobility for centuries**. Many **Mughal paintings** show princes and emperors enjoying a hunt." — but "under colonial rule **the scale of hunting increased to such an extent that various species became almost extinct**."*

The colonial justification — and it connects straight back to Chapter 3's misuse of "civilising":

★ ★ ★ *"The British saw large animals as signs of a wild, primitive and savage society. They believed that by killing dangerous animals the British would civilise India. They gave **rewards for the killing of tigers, wolves and other large animals** on the grounds that **they posed a threat to cultivators**."*

The numbers — ★ ★ ★ learn these:

Killed	Figure
Tigers	over 80,000)
Leopards	150,000 { killed for reward, 1875–1925
Wolves	200,000)
★ The Maharaja of Sarguja alone	1,157 tigers and 2,000 leopards up to 1957
★ George Yule, a British administrator	400 tigers

The end of the story: *"Gradually, the tiger came to be seen as **a sporting trophy**. Initially **certain areas of forests were reserved for hunting**. **Only much later** did environmentalists and conservators begin to argue that **all these species of animals needed to be protected, and not killed**."*

★ ★ ★ ★ The perfect illustration of the double standard, and it makes an excellent short answer: a Baiga hunting a partridge for food became a poacher; a Viceroy photographed behind a row of dead tigers was a sportsman. NCERT even tells you who did the actual work: *"the tracking was done by **skilled village hunters**, and the **Sahib simply fired the shot**."*

🛍️ 2.4 New trades, new employments, new services

Some gained:

*"While people **lost out in many ways** after the forest department took control of the forests, **some people BENEFITED from the new opportunities that had opened up in trade**. Many communities left their traditional occupations and started trading in forest products."*

Case	Detail
Brazil	With the growing demand for rubber in the mid-nineteenth century, the ★ Mundurucu peoples of the Brazilian Amazon — who lived in villages on high ground and cultivated manioc — <i>"began to collect latex from wild rubber trees for supplying to traders. Gradually, they descended to live in trading posts and became COMPLETELY DEPENDENT on traders"</i>
India — before the British	<i>"the trade in forest products was not new. From the medieval period onwards, we have records of adivasi communities trading elephants and other goods like hides, horns, silk cocoons, ivory, bamboo, spices, fibres, grasses, gums and resins through nomadic communities like the BANJARAS"</i>
★★ India — under the British	<i>"trade was completely regulated by the government. The British government gave many large European trading firms the SOLE RIGHT to trade in the forest products of particular areas. Grazing and hunting by local people were restricted."*</i>

Who lost:

★★★ *"In the process, many **pastoralist and nomadic communities like the KORAVA, KARACHA and YERUKULA of the Madras Presidency** lost their livelihoods. Some of them began to be called '**CRIMINAL TRIBES**', and were forced to work instead in **FACTORIES, MINES and PLANTATIONS**, under government supervision."*

★★★★ *"New opportunities of work did not always mean improved well-being for the people. In **Assam**, both men and women from forest communities like ★ **SANTHALS and ORAONS from Jharkhand**, and **GONDS from Chhattisgarh** were recruited to work on **TEA PLANTATIONS**. Their **wages were low** and **conditions of work were very bad**. They could not return easily to their home villages from where they had been recruited."*

Source C — the Baigas of Central India, petitioning in 1892 after their shifting cultivation was stopped. Read it whole; it is the human centre of the chapter: *"**We daily starve, having had no foodgrain in our possession. The only wealth we possess is our axe.** We have no clothes to cover our body with, but we pass cold nights by the fireside. We are now dying for want of food. We cannot go elsewhere. **What fault have we done that the government does not take care of us? PRISONERS ARE SUPPLIED WITH AMPLE FOOD IN JAIL. A cultivator of the grass is not deprived of his holding, but the government does not give us our right who have lived here for generations past.**"* 🌀 *"Prisoners are supplied with ample food in jail" — the Baigas are pointing out that the state feeds those it punishes and starves those it has merely dispossessed.* Quote this in any answer on colonial dispossession.

Source D — the Putumayo, for scale of what plantation extraction could mean: rubber extraction in the Amazon by the **Peruvian Rubber Company** (with British and Peruvian interests) depended on **the forced labour of the local Huitotos**. ★ *"From 1900–1912, the Putumayo output of 4,000 tons of rubber was associated with a decrease of some 30,000 among the Indian population due to torture, disease and flight."*

3 Rebellion in the Forest — Bastar, 1910

*"In many parts of India, and across the world, **forest communities rebelled** against the changes that were being imposed on them."*

The leaders NCERT names — *"still remembered today in songs and stories"*.

Leader	Place
★ Siddhu and Kanu	the Santhal Parganas
★★ Birsa Munda	Chhotanagpur (you know him already — Class 8 Ch.4)
★★ Alluri Sitarama Raju	Andhra Pradesh

3.1 The people of Bastar

Feature	Detail
Location	"the southernmost part of Chhattisgarh , bordering Andhra Pradesh, Orissa and Maharashtra "
Geography	The central part is a plateau ; to its north the Chhattisgarh plain , to its south the Godavari plain ; ★ the river Indrawati winds across Bastar east to west
★ Communities	Maria and Muria Gonds, Dhurwas, Bhatras and Halbas — "They speak different languages but share common customs and beliefs "

Their system of land and forest — ★★ ★ and note that this is a functioning conservation regime:

THE BELIEF

"each village was GIVEN ITS LAND BY THE EARTH, and in return, they LOOK AFTER THE EARTH by making some offerings at each agricultural festival."

They also "show respect to the SPIRITS OF THE RIVER, THE FOREST AND THE MOUNTAIN."

THE PRACTICE

- "Since each village KNOWS WHERE ITS BOUNDARIES LIE, the local people LOOK AFTER ALL THE NATURAL RESOURCES within that boundary."
- If people from one village want wood from ANOTHER village's forest, they pay a small fee -- DEVSARI, DAND or MAN.
- "Some villages also PROTECT THEIR FORESTS BY ENGAGING WATCHMEN, and EACH HOUSEHOLD CONTRIBUTES SOME GRAIN to pay them."
- "Every year there is ONE BIG HUNT where the HEADMEN of villages in a PARGANA (cluster of villages) meet and DISCUSS ISSUES OF CONCERN, INCLUDING FORESTS."

★★★★ This paragraph is doing serious analytical work, and most students read past it. Bastar already had **defined boundaries, a transfer fee for cross-village use, paid watchmen funded by household contribution, and an annual multi-village assembly to deliberate on forest matters**. In other words, **a property regime, a pricing mechanism, an enforcement service and a governance forum**. The colonial claim that forests needed "scientific management" because locals could not manage them is refuted by the chapter's own evidence. **This is exactly the argument that returns in the 1980s and in the Forest Rights Act.**

🧑 3.2 The fears of the people, and the rebellion

1905 -- The colonial government proposes to RESERVE TWO-THIRDS OF THE FOREST, and to STOP shifting cultivation, hunting and collection of forest produce.

Some villages were allowed to STAY ON in the reserved forests -- ON THE CONDITION that they "WORKED FREE FOR THE FOREST DEPARTMENT in CUTTING AND TRANSPORTING TREES, and PROTECTING THE FOREST FROM FIRES."

"Subsequently, these came to be known as 'FOREST VILLAGES'."

"People of other villages were DISPLACED WITHOUT ANY NOTICE OR COMPENSATION."

THE ACCUMULATED GRIEVANCES

- "increased LAND RENTS"
- "frequent demands for FREE LABOUR AND GOODS by colonial officials"
- THE TERRIBLE FAMINES: 1899-1900 and again 1907-1908

"RESERVATIONS PROVED TO BE THE LAST STRAW."

How the rebellion was organised — this is the memorable part:

*"People began to gather and discuss these issues **in their village councils, in bazaars and at festivals** or wherever the **headmen and priests of several villages were assembled**. The initiative was taken by the ★ **DHURWAS of the KANGER forest, where reservation first took place**. Although there **was NO SINGLE LEADER**, many people speak of ★★ **GUNDA DHUR, from village NETHANAR**, as an important figure in the movement."*

★★★★★ *"In 1910, **MANGO BOUGHS, A LUMP OF EARTH, CHILLIES AND ARROWS** began circulating between villages. **These were actually MESSAGES INVITING VILLAGERS TO REBEL against the British**. Every village contributed something to the rebellion expenses."*

What they did — and note the precision of the targets:

*"**Bazaars were looted, the houses of officials and traders, schools and police stations were burnt and robbed, and grain redistributed.** ★★ Most of those who were attacked were **IN SOME WAY ASSOCIATED WITH THE COLONIAL STATE AND ITS OPPRESSIVE LAWS.**"*
William Ward, a missionary who observed the events: "From all directions came streaming into Jagdalpur, police, merchants, forest peons, schoolmasters and immigrants."

🎯 **That list of who fled is itself the answer to "who did the rebels attack?" — police, merchants, forest peons, schoolmasters, immigrants. Not random violence: a precise strike at the personnel of the colonial order.**

The suppression:

"The British sent TROOPS to suppress the rebellion.
The adivasi leaders TRIED TO NEGOTIATE, but THE BRITISH SURROUNDED THEIR CAMPS AND FIRED UPON THEM.
After that they MARCHED THROUGH THE VILLAGES FLOGGING AND PUNISHING those who had taken part.
MOST VILLAGES WERE DESERTED as people fled into the jungles.
It took THREE MONTHS (FEBRUARY – MAY) for the British to regain control.
HOWEVER, THEY NEVER MANAGED TO CAPTURE GUNDA DHUR."

★★★★ THE OUTCOME — and this is the point of telling the story: *"In a MAJOR VICTORY FOR THE REBELS, work on reservation was TEMPORARILY SUSPENDED, and the area to be reserved was REDUCED TO ROUGHLY HALF of that planned before 1910."* The rebellion was **militarily crushed and politically successful**. That distinction is worth making explicitly in an answer — colonial policy was **repeatedly modified by resistance**, which is why NCERT says forest dwellers *"also rebelled against the rules and persuaded the state to change its policies."*

Source F — the memory of the battle, recounted by elders from their parents:

Podiyami Ganga of Kankapal, quoting his father: *"The British came and started taking land. The Raja didn't pay attention... his supporters gathered people. War started. His staunch supporters died and the rest were whipped. My father... suffered many strokes, but he escaped and survived. It was a movement to get rid of the British. The British used to tie them to horses and pull them."*

Chendru, of village Nandrasa: *"People from every pargana camped in Alnar tarai. The paltan (force) surrounded the people in a flash. Gunda Dhur had flying powers and flew away. BUT WHAT COULD THOSE WITH BOWS AND ARROWS DO? The battle took place at night. The people hid in shrubs and crawled away..."*

🎯★★★★ Two things to notice about these sources. First, this is **oral history** — grandchildren repeating what parents were told, which is often the **only** archive a non-literate community leaves. (Recall **Class 8 Ch.1**: official records tell us what officials thought, not what others felt. **This is what the other side sounds like.**) Second, *"Gunda Dhur had flying powers and flew away"* is not a factual error — **it is how a community remembers a leader the British never caught**. Myth here is a record of an actual outcome.

🌳 **And after Independence — do not skip this paragraph**

★★★★★ *"The story of the forests and people of Bastar does not end there. After Independence, the SAME PRACTICE of keeping people out of the forests and reserving them for INDUSTRIAL USE CONTINUED. In the 1970s, the WORLD BANK proposed that 4,600 hectares of natural SAL forest should be replaced by TROPICAL PINE to provide PULP FOR THE PAPER INDUSTRY. It was only after protests by local environmentalists that the project was stopped."*

★★★★★ This is the most important sentence in the chapter for a Mains answer. It refuses the comfortable conclusion that the problem was *colonialism*. **The independent Indian state inherited the Forest Department, the Acts, the categories and the logic — and in the 1970s was still proposing to replace a natural sal forest with a pine monoculture for paper pulp.** *Scientific forestry* outlived the empire that invented it. Any answer that ends in 1947 has missed the point NCERT is making.

4 Forest Transformations in Java

Why NCERT sends you to Indonesia: to show that this was **not a British peculiarity but a colonial pattern**. *"The colonial power in Indonesia were the Dutch, and... there were many similarities in the laws for forest control in Indonesia and India."*

Fact	Detail
Java today	"famous as a rice-producing island " — but "once upon a time it was covered mostly with forests "
★ Why the Dutch wanted it	"Like the British, they wanted timber from Java to build ships "
★ Population of Java, 1600	an estimated 3.4 million — many villages in the fertile plains, " but also many communities living in the mountains and practising shifting cultivation "

🔪 4.1 The Kalangs — the woodcutters of Java

*"The ★ **KALANGS** of Java were a community of **skilled forest cutters and shifting cultivators**. They were so valuable that in ★ **1755**, when the **MATARAM** kingdom of Java split, the **6,000 KALANG FAMILIES WERE EQUALLY DIVIDED BETWEEN THE TWO KINGDOMS**. Without their expertise, it would have been difficult to **harvest teak** and **for the kings to build their palaces**. When the Dutch began to gain control over the forests in the eighteenth century, they tried to make the Kalangs work under them. ★ **In 1770, the Kalangs RESISTED by attacking a Dutch fort at JOANA, but the uprising was suppressed.**"*

🎯 **A community divided like property between two kingdoms** — that detail alone tells you what skilled forest labour was worth, and it makes an unforgettable exam fact.

⚖️ 4.2 Dutch scientific forestry

★★★★★ "In the nineteenth century, when it became important to **CONTROL TERRITORY AND NOT JUST PEOPLE**, the Dutch enacted **forest laws in Java, restricting villagers' access to forests.**"

That phrase — **territory and not just people** — is the sharpest single formulation of what changes under modern colonial rule, and it applies equally to India. Earlier rulers wanted **labour, tribute and loyalty from persons**. The colonial state wanted **the land itself, surveyed, classified and owned**.

Restriction	Detail
Permitted use	Wood could only be cut for specified purposes like making river boats or constructing houses , and only from specific forests under close supervision
Punishable offences	" grazing cattle in young stands, transporting wood without a permit, or travelling on forest roads with horse carts or cattle "
★ Sleepers exported from Java, 1882	280,000

And the labour system — ★★★★★ learn the name:

The problem: "all this required **LABOUR** to **CUT** the trees, **TRANSPORT** the logs and **PREPARE** the sleepers."

STEP 1 The Dutch first **IMPOSED RENTS** on land being cultivated in the forest.

STEP 2 Then **EXEMPTED SOME VILLAGES** from these rents -- IF they "**worked COLLECTIVELY** to provide **FREE LABOUR AND BUFFALOES** for cutting and transporting timber."

THIS WAS KNOWN AS THE *****BLANDONGDIENSTEN***** SYSTEM.
STEP 3 "LATER, instead of rent exemption, forest villagers were **GIVEN SMALL WAGES**, but **THEIR RIGHT TO CULTIVATE FOREST LAND WAS RESTRICTED.**"

★ ★ ★ Compare with Bastar's "forest villages" — where villagers could stay in the reserved forest on condition of working free for the forest department. Two empires, two continents, one device: create a tax or a prohibition, then remit it in exchange for unpaid labour. That is the single best answer to NCERT's Question 2 on the similarities between Bastar and Java.

👉 4.3 Samin's challenge

"Around 1890, ★ ★ SURONTIKO SAMIN of RANDUBLATUNG village, a teak forest village, began questioning state ownership of the forest. ★ ★ ★ He argued that the state had NOT CREATED the wind, water, earth and wood, SO IT COULD NOT OWN IT."

Spread	"Soon a widespread movement developed. Amongst those who helped organise it were Samin's sons-in-law "
★ By 1907	3,000 families were following his ideas
★ ★ The methods	"Some of the Saminists protested by LYING DOWN ON THEIR LAND when the Dutch came to survey it , while others refused to pay taxes or fines or perform labour "

★ ★ ★ ★ Samin's argument is a philosophical one, not merely a grievance, and it is worth carrying into an Essay. He does not say the tax is too high or the rule unfair; he denies the state's title. Ownership must be founded on creation or on labour — the state did neither, therefore it owns nothing. And the method matches the argument: **lying down on the land when the surveyors came** is a refusal to be mapped, which is precisely how the colonial state converted land into property. **Non-cooperation with the survey is non-cooperation with the claim.** (Compare Gandhi's methods, three decades later and a sea away.)

👉 4.4 War and deforestation — ★ ★ ★ NCERT asks this as a direct question

IN INDIA

"The First World War and the Second World War had a MAJOR IMPACT on forests. In India, WORKING PLANS WERE ABANDONED at this time, and THE FOREST DEPARTMENT CUT TREES FREELY TO MEET BRITISH WAR NEEDS."

IN JAVA

"Just before the Japanese occupied the region, the DUTCH followed a 'SCORCHED EARTH' POLICY, DESTROYING SAWMILLS and BURNING HUGE PILES OF GIANT TEAK LOGS so that they would not fall into Japanese hands."

"The JAPANESE then EXPLOITED THE FORESTS RECKLESSLY for their own war industries, FORCING FOREST VILLAGERS TO CUT DOWN FORESTS."

"MANY VILLAGERS USED THIS OPPORTUNITY TO EXPAND CULTIVATION IN THE FOREST. AFTER THE WAR, IT WAS DIFFICULT FOR THE INDONESIAN FOREST SERVICE TO GET THIS LAND BACK."

"As in India, PEOPLE'S NEED FOR AGRICULTURAL LAND has brought them into CONFLICT WITH THE FOREST DEPARTMENT'S DESIRE TO CONTROL THE LAND AND EXCLUDE PEOPLE FROM IT."

★ ★ ★ So "why are forests affected by wars?" has **THREE distinct mechanisms**, and a full answer gives all three: **(i) surging demand** — war needs timber, so conservation plans are abandoned and cutting becomes unrestrained; **(ii) deliberate destruction** — scorched-earth denial of resources to the enemy; **(iii) the collapse of state control** — when the forest department's grip loosens, **villagers re-occupy the land**, and the post-war state cannot get it back. **The third is the most interesting, because it shows that forest "protection" was always a matter of enforcement capacity, not consent.** And note the caption to Fig. 23: **"The Allies would not have been as successful in the First World War and the Second World War if they had not been able to exploit the resources and people of their colonies."** — a directly quotable line for any answer on **colonies and the world wars**.

🌱 4.5 New developments in forestry — ★ ★ ★ ★ the GS3 payoff

*"Since the 1980s, governments across Asia and Africa have begun to see that **scientific forestry and the policy of keeping forest communities away from forests has RESULTED IN MANY CONFLICTS.** ★ ★ CONSERVATION of forests rather than COLLECTING TIMBER has become a more important goal. The government has recognised that in order to meet this goal, **the people who live near the forests MUST BE INVOLVED.**"*

Evidence NCERT gives	Detail
★ ★ ★ Sacred groves	"In many cases, across India, from MIZORAM to KERALA, dense forests have survived ONLY because villages protected them in SACRED GROVES known as SARNAS, DEVARAKUDU, KAN, RAI, etc. "
★ ★ Community patrolling	"Some villages have been patrolling their own forests, with each household taking it in turns , instead of leaving it to the forest guards"
The present	"Local forest communities and environmentalists today are thinking of different forms of forest management "

★ ★ ★ ★ Close the circle when you answer. The village watchmen of **Bastar**, paid in grain by each household in 1910, and the villages **patrolling their own forests by household turns** today, are **the same institution**. A century of "scientific" management ended by re-adopting the arrangement it had displaced. **For GS3, connect this to: Joint Forest Management (1990), the Forest Rights Act 2006, PESA 1996, and the biodiversity case against monoculture plantations.**

 Prelims traps

Trap	The truth
"Brandis was British"	★ A GERMAN expert , invited for advice and made the first Inspector General of Forests in India
"The Indian Forest Service was set up in 1865"	★★ Indian Forest Service — 1864. Indian Forest Act — 1865. Amended 1878 and 1927
"The Forest Research Institute was set up at Dehradun in 1864"	★ 1906 (the Imperial Forest Research Institute); the forest school there was the first in the British Empire
"The 1865 Act created the three categories"	★★ The 1878 amendment divided forests into reserved, protected and village
"Villagers could take fuelwood from reserved forests"	★★ Nothing, even for their own use. Wood for house building or fuel came from protected or village forests
"Scientific forestry means conserving mixed forests"	★★ It means cutting natural mixed forest and planting ONE species in straight rows — a plantation . NCERT notes ecologists say it is " not scientific at all "
"Jhum is a Sri Lankan term"	★ Chena = Sri Lanka; lading = SE Asia; milpa = Central America; chitemene/tavy = Africa. Indian: dhya, penda, bewar, nevad, jhum , podu, khandad, kumri
"Shifting-cultivation plots are left fallow for 2–3 years"	★★ 12 to 18 years
"The British banned shifting cultivation only for ecological reasons"	★★ Three reasons — land could not grow railway timber, fire risk to valuable timber, and it made tax calculation harder
"Each mile of track needed about 200 sleepers"	★★ 1,760 to 2,000
"25,500 km of track by 1946"	★ 25,500 km by 1890; over 765,000 km by 1946
"80,000 tigers were killed between 1875 and 1925"	★ Over 80,000 tigers, 150,000 leopards and 200,000 wolves killed for reward in that period
"The Bastar rebellion was in 1905"	★★ Reservation was proposed in 1905; the rebellion was in 1910
"Gunda Dhur was captured and hanged"	★★ ***They never managed to capture Gunda Dhur.*** And reservation work was suspended and the reserved area halved
"Mango boughs and chillies were religious offerings"	★★ Mango boughs, a lump of earth, chillies and arrows were messages inviting villagers to rebel
"Blandongdiensten was an Indian labour system"	★★ Dutch, in Java — rent exemption in exchange for free labour and buffaloes
"Samin led an armed revolt"	★ Saminists lay down on their land when surveyors came, and refused taxes, fines and labour
"The Kalangs were divided between two kingdoms in 1755"	✓ True — 6,000 Kalang families, equally divided when the Mataram kingdom split. They attacked the Dutch fort at Joana in 1770
"Terra nullius was a legal fact in Australia"	★ A claim that the continent was empty — while settlers were " guided through the landscape by aboriginal tracks, and led by aboriginal guides "
"Criminal tribes were so labelled for crimes committed"	★★ Korava, Karacha and Yerukula were so labelled after losing their livelihoods to trade restrictions, then forced into factories, mines and plantations under government supervision

● Mains angles

- ★★★★★ "Discuss how changes in forest management in the colonial period affected: shifting cultivators; nomadic and pastoralist communities; firms trading in timber/forest produce; plantation owners; kings and British officials engaged in shikar." (NCERT Q1 — write it as a ledger of winners and losers.)

Group	Effect
Shifting cultivators	Practice banned ; forcibly displaced ; forced to change occupation; some rebelled — the Baigas' 1892 petition is the evidence
Nomadic and pastoralist communities	Grazing and hunting restricted ; Korava, Karacha, Yerukula lost livelihoods ; branded " criminal tribes "; pushed into factories, mines, plantations
Firms trading in timber/forest produce	Gained — large European firms received the SOLE RIGHT to trade in particular areas
Plantation owners	Gained — forests taken over by the state and handed to European planters at cheap rates
Kings / British officials engaged in shikar	Gained — customary hunting became poaching for villagers, while big-game hunting became sport , with rewards and reserved hunting areas

- ★★★★★ "What are the similarities between colonial management of the forests in Bastar and in Java?" (NCERT Q2.) — (i) Both colonial states wanted **timber for ships and railways**; (ii) both enacted **forest laws restricting villagers' access** and criminalised **everyday use** — grazing, wood-cutting, transport without permit; (iii) both **banned or squeezed shifting cultivation**; (iv) both created a **forest service and "scientific forestry" plantations**; (v) ★ both extracted **unpaid labour by letting villagers stay in exchange for free work** — **forest villages** in Bastar, *blandongdiensten* in Java; (vi) both provoked **resistance** — Bastar 1910, the **Kalangs 1770** and the **Saminists from 1890**; (vii) both were driven by the shift to **controlling territory rather than just people**.
- ★★★★★ "Between 1880 and 1920 forest cover in the Indian subcontinent declined by 9.7 million hectares, from 108.6 to 98.9 million hectares. Discuss the role of: railways, shipbuilding, agricultural expansion, commercial farming, tea/coffee plantations, adivasis and other peasant users." (NCERT Q3 — note that the last item is a trap: *adivasi use was the smallest driver and the most blamed. Say so.*)
- ★★★★★ "Why are forests affected by wars?" (NCERT Q4 — the three mechanisms above.)
- ★★★★★ "Scientific forestry was systematic but not scientific." **Examine.** — Monoculture vs 500 species per patch; single-output optimisation; loss of fodder, fuel, food, medicine, fibre; and the ecological objections ecologists raise today.
- ★★★★★ **Bastar 1910 as a case study in colonial dispossession and resistance** — the pre-existing conservation regime, the 1905 reservation proposal, the famines, the mango bough and chillies, the precise targeting of colonial personnel, the suppression, and **the political victory** of halved reservation.
- ★★★★★ **GS3 crossover: "Community-based forest management is not a new idea but a recovered one." Discuss.** — Bastar's **devsari** fee, boundary knowledge, grain-paid watchmen and pargana assemblies; **sacred groves** (sarna, devarakudu, kan, rai) from **Mizoram to Kerala**; household-turn patrolling; then **JFM, PESA 1996 and the Forest Rights Act 2006. This is the single most valuable answer you can build from this chapter.**
- ★★★★★ "Colonial forest policy did not end in 1947." — The World Bank's **1970s proposal to replace 4,600 hectares of natural sal with tropical pine** for paper pulp, stopped only by local environmentalists.

🔗 Cross-links to build now

- ★★★★★ **Class 8 Ch.4** — the **same** forest policy from the tribal side: forests declared state property, reserved forests barring jhum/fruit/hunting, **forest villages created for cheap labour**, and **Birsa Munda**, who

appears in **both** chapters.

- **Ch.5 of this book** — the **Waste Land Rules**, grazing bans and the **Criminal Tribes Act 1871** apply the identical logic to people who move.
- **Ch.2 of this book** — the Bolsheviks' "*harsh discouragement of nomadism*" in Central Asia: **the same suspicion of mobile people from the opposite ideology**.
- **Class 8 Ch.1** — official records vs oral testimony; **Source F from Bastar is what the other archive sounds like**.
- **GS3** — Forest Rights Act 2006, JFM, PESA, sacred groves, biodiversity vs monoculture, the Western Ghats.

Glossary

Term	Meaning
Deforestation	The disappearance of forests
Terra nullius	" <i>Empty land</i> " — the settler claim that a continent was uninhabited
Sleepers	Wooden planks laid across railway tracks; they hold the tracks in position
Scientific forestry	A system of cutting trees controlled by the forest department , in which old trees are cut and new ones planted
Plantation	Natural mixed forest replaced by one type of tree planted in straight rows
Working plan	The forest department's plan for how much of the plantation area to cut each year
Reserved forest	The best forests — villagers could take nothing , even for their own use
Protected / village forest	Forests from which wood could be taken for house building or fuel
Shifting cultivation / swidden	Parts of the forest cut and burnt in rotation , seeds sown in the ashes, plots left fallow 12–18 years
Jhum, podu, bewar, dhya, penda, nevad, khandad, kumri	Indian names for shifting cultivation
Lading / milpa / chitemene / tavy / chena	SE Asia / Central America / Africa / Africa / Sri Lanka
Poaching	The offence customary hunting became after the forest laws
Forest villages	Villages allowed to remain in reserved forests on condition of working free for the forest department
Blandongdiensten	The Dutch Javanese system — rent exemption in exchange for free labour and buffaloes
Devsari, dand, man	The fee paid in Bastar for taking wood from another village's forest
Pargana	A cluster of villages , whose headmen met annually at the big hunt
Criminal tribes	Communities such as Korava, Karacha and Yerukula , so labelled after losing their livelihoods
Taungya	The system allowing local farmers to cultivate temporarily within a plantation
Sarna, devarakudu, kan, rai	Names for sacred groves protected by villages
Banjaras	Nomadic trading communities through whom adivasi forest goods moved from the medieval period
Mahua (Madhuca indica)	Tree whose flowers are eaten or made into alcohol and whose seeds and fruit give oil
Tendu	Leaves collected in bundles of ~50, a major source of forest income

60-Second Recap

1. **Deforestation is old, but "under colonial rule it became more SYSTEMATIC and EXTENSIVE."**
★ **13.9 million sq km — 9.3% of the world's land — cleared between 1700 and 1995.**
2. **Cause 1 — cultivation.** India went from **one-sixth** under the plough in **1600** to about **half** today; **cultivated area rose 6.7 million hectares between 1880 and 1920**, driven by **commercial crops (jute, sugar, wheat, cotton)** and by the belief that **forests were "unproductive wilderness"** that should yield **revenue**.
3. ★ **Terra nullius** — the settler claim that land was empty. In Australia the settlers were **guided by aboriginal tracks and aboriginal guides**; the **Ngarrindjeri** mapped their land on the body of the ancestor **Ngurunderi**. **"It was an argument that justified conquest"** — and in Honduras it was followed three years later by the **United Fruit Company** and the **"Banana Republics."**
4. **Cause 2 — ships.** England's **oaks were disappearing by the early 19th century**; **search parties went to India by the 1820s**, and within a decade timber was being **exported on a massive scale**.
5. ★ **Cause 3 — railways, from the 1850s.** **1,760–2,000 sleepers per mile**; **25,500 km of track by 1890**, over **765,000 km by 1946**; **35,000 trees a year in the Madras Presidency alone by the 1850s**; and **contractors cutting indiscriminately**.
6. **Cause 4 — plantations.** Forests cleared for **tea, coffee and rubber**, handed to **European planters at cheap rates**.
7. ★★ **Dietrich Brandis**, a **German**, became the **first Inspector General of Forests**; set up the **Indian Forest Service (1864)**, helped frame the **Indian Forest Act (1865)**; the **Imperial Forest Research Institute** came up at **Dehradun in 1906**.
8. ★★ **"Scientific forestry"** replaced mixed natural forest with **single-species plantations in straight rows** — *"Many people now, including ecologists, feel that this system is not scientific at all."*
9. ★★ **The 1878 amendment** created **reserved** (villagers could take **nothing**), **protected** and **village** forests.
10. ★★ **Foresters wanted tall, straight hardwood — teak and sal — for ships and railways**; **villagers wanted a mixture of species for fuel, fodder and leaves**. After the Act, **all everyday practices became illegal**, people were **forced to steal wood**, and forest guards **took bribes** and demanded **free food**.
11. ★★ **Shifting cultivation** — burnt in rotation, sown in the ashes after the first monsoon, harvested by **October–November**, fallow **12–18 years**. Banned because the land **could not grow railway timber**, because **fire threatened valuable timber**, and because it **made taxes hard to calculate**.
12. ★★ **Hunting**: customary hunting became **poaching**, while **big game became sport** — over **80,000 tigers**, **150,000 leopards** and **200,000 wolves killed for reward, 1875–1925**; the **Maharaja of Sarguja** shot **1,157 tigers**; **George Yule 400**. Justification: killing dangerous animals would **"civilise India."**
13. **Trade**: the **Mundurucu** of the Amazon became **dependent on rubber traders**; in India, trade that had run through **Banjaras** since medieval times was **monopolised by large European firms**; **Korava, Karacha and Yerukula** lost livelihoods and were branded **"criminal tribes"**; **Santhals, Oraons and Gonds** were recruited to **Assam tea plantations** at low wages and **could not easily return home**. The **Baigas' 1892 petition**: *"Prisoners are supplied with ample food in jail."*
14. ★★ **Bastar 1910**. The people already had **village boundaries, the devsari fee, grain-paid watchmen and an annual pargana assembly on forest matters**. In **1905** the state proposed to **reserve two-thirds of the forest**; some villages could stay only by **working free** ("forest villages"), others were **displaced without notice or compensation**; then came **famines in 1899–1900 and 1907–08**. *"Reservations proved to be the last straw."*

15. ★★ **Mango boughs, a lump of earth, chillies and arrows** circulated as **calls to rebel**; every village contributed to the expenses; the **Dhurwas of the Kanger forest** began it and **Gunda Dhur of Nethanar** is remembered, **though there was no single leader**. Targets were **officials, traders, schools and police stations**. The British **fired on negotiating camps** and took **three months (February–May)** to regain control — but ★ **never caught Gunda Dhur**, and **reservation was suspended and the reserved area halved**.
16. ★★ **And it did not end in 1947**: in the **1970s the World Bank proposed replacing 4,600 hectares of natural sal with tropical pine** for paper pulp, stopped **only by local environmentalists**.
17. ★ **Java**: the Dutch wanted **timber for ships**; the **Kalangs**, skilled cutters, were **so valuable that 6,000 families were split equally when the Mataram kingdom divided in 1755**, and they **attacked the Dutch fort at Joana in 1770**. In the 19th century, "*when it became important to **control territory and not just people***," forest laws restricted access — punishing **grazing in young stands, transport without a permit, and travelling forest roads with carts or cattle**. **280,000 sleepers were exported from Java in 1882**. Labour came through ★ **blandongdiensten** — **rent exemption for free labour and buffaloes**.
18. ★★ **Samin, from 1890**: "*the state had not created the wind, water, earth and wood, so it could not own it*." **3,000 families by 1907**; Saminists **lay down on their land when surveyors came and refused taxes, fines and labour**.
19. ★ **Wars**: working plans **abandoned** in India and trees cut freely for British war needs; the Dutch **scorched-earth** burning of teak before the Japanese arrived; the **Japanese exploited the forests recklessly**; and **villagers expanded cultivation into the forest, land the forest service could not recover afterwards**.
20. ★★ **Since the 1980s**: conservation has displaced timber collection as the goal, and the state has recognised that **people who live near forests must be involved**. Dense forests survive **from Mizoram to Kerala in sacred groves** — **sarna, devarakudu, kan, rai** — and villages **patrol their own forests by household turns**.

Test Yourself

Prelims style

1. Match: Indian Forest Service / Indian Forest Act / Forest Research Institute, Dehradun. — **1864 / 1865 / 1906**.
2. The 1878 amendment to the Forest Act divided forests into — **reserved, protected and village forests**; villagers could take **nothing at all** from **reserved** forests.
3. *Milpa, chena, chitemene, kumri* — which is Indian? — **Kumri**. (*milpa* = Central America, *chena* = Sri Lanka, *chitemene* = Africa.)
4. Consider: (i) Shifting-cultivation plots are left fallow for 12 to 18 years. (ii) The British banned shifting cultivation partly because it made tax calculation harder. (iii) Dietrich Brandis was a British forester. — **(i) and (ii) only**. Brandis was **German**.
5. Each mile of railway track required — **between 1,760 and 2,000 sleepers**; by **1890** about **25,500 km** of track had been laid, rising to over **765,000 km** by **1946**.
6. The *blandongdiensten* system refers to — **the Dutch system in Java of exempting villages from rent in exchange for free labour and buffaloes for cutting and transporting timber**.
7. Gunda Dhur is associated with — **the Bastar rebellion of 1910**; reservation was proposed in **1905**, and after the rebellion the area to be reserved was **halved**.

8. Sarna, devarakudu, kan and rai are — ***names for sacred groves protected by villages.***

Mains style

9. Discuss how changes in forest management in the colonial period affected shifting cultivators, nomadic and pastoralist communities, timber firms, plantation owners, and those engaged in shikar. **(250 words)** ★
10. What are the similarities between colonial management of forests in Bastar and in Java? **(250 words)** ★
11. Discuss the role of railways, shipbuilding, agricultural expansion, commercial farming, plantations and adivasi users in the decline of forest cover between 1880 and 1920. **(250 words)** ★
12. Why are forests affected by wars? **(150 words)**
13. "*Scientific forestry was systematic, but not scientific.*" Examine. **(250 words)** ★
14. "*Community-based forest management in India is not a new idea but a recovered one.*" Discuss with reference to colonial forest policy and post-1980s developments. **(250 words)** ★ — *the GS3 crossover*

Self-check (out loud, no notes):

- Give the four causes of colonial deforestation, with one figure for each.
- Explain *terra nullius* and why it mattered.
- Name Brandis's three institutions with their years.
- Define scientific forestry and give two reasons ecologists dispute the name.
- Distinguish the three forest categories.
- Describe shifting cultivation and give the three reasons it was banned.
- Give the hunting figures and explain the double standard.
- Narrate Bastar 1910 from the 1905 proposal to the halved reservation.
- List five similarities between Bastar and Java.
- Give the three mechanisms by which war destroys forests.